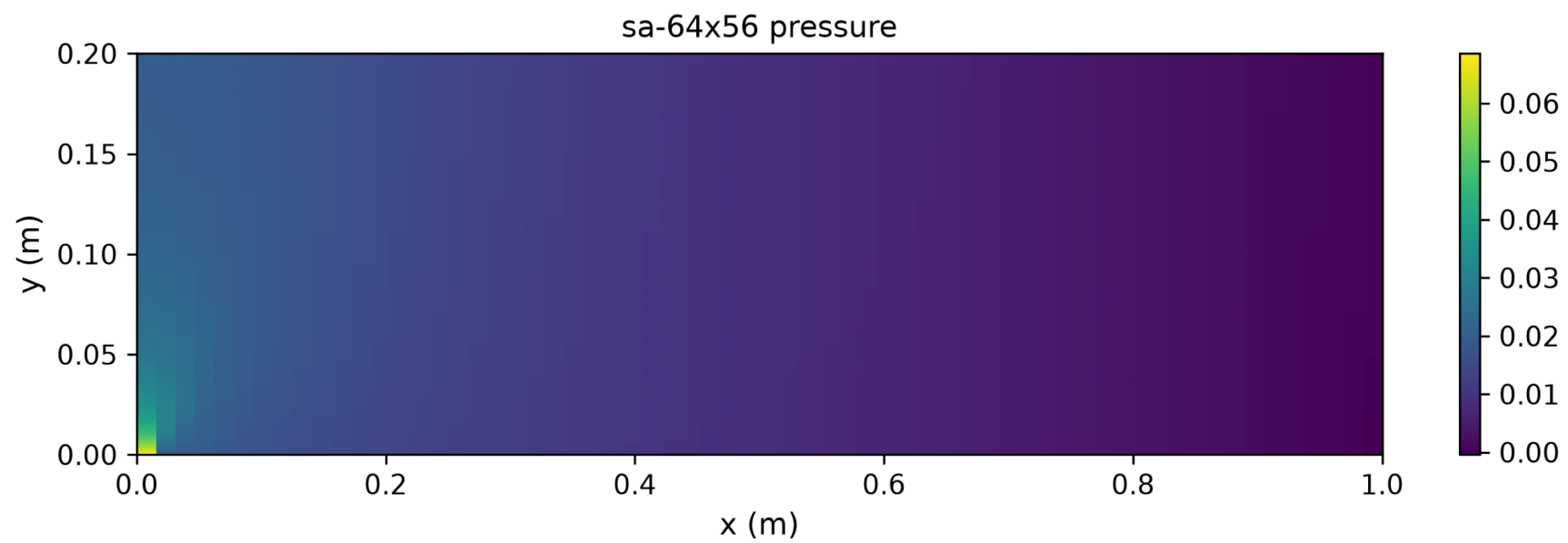
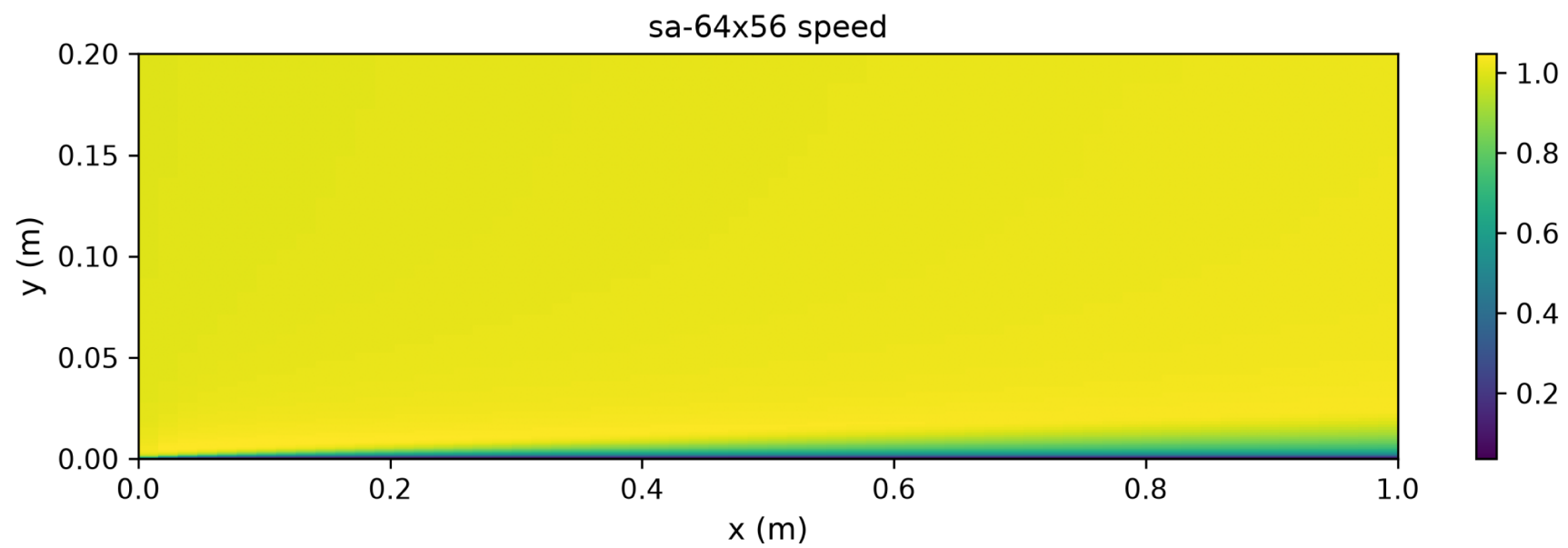


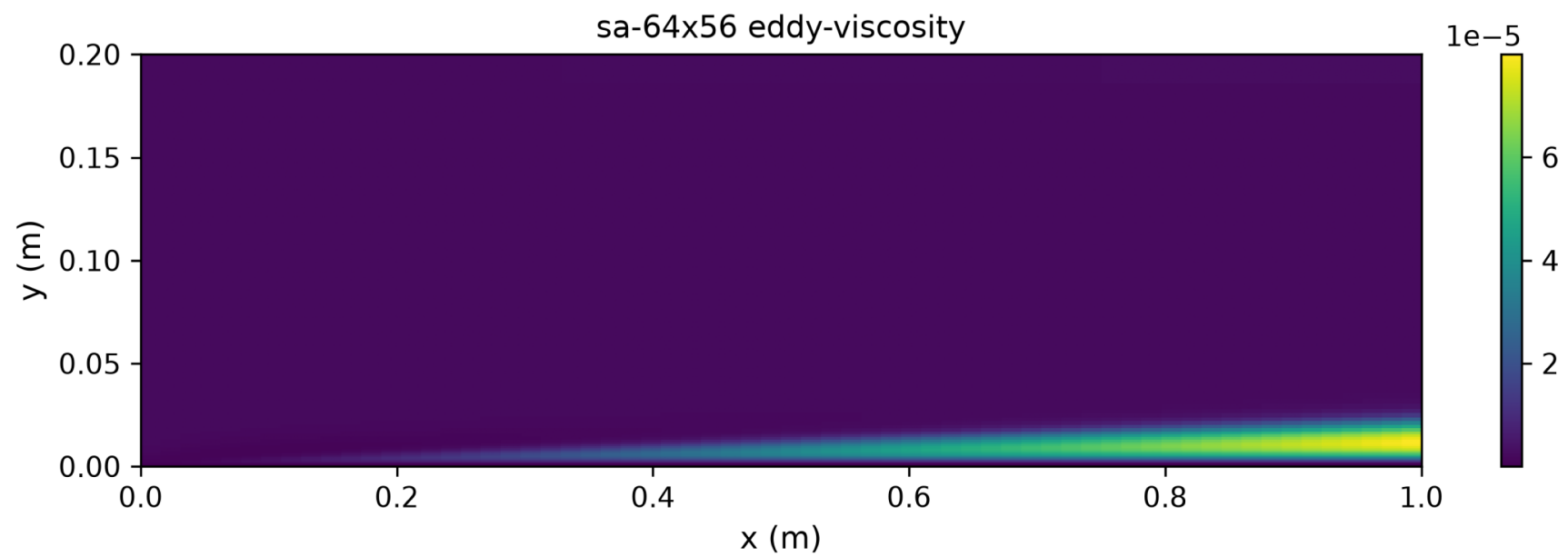
SA mesh refinement, with actual CPU/CUDA devices: convergence and empirical accuracy

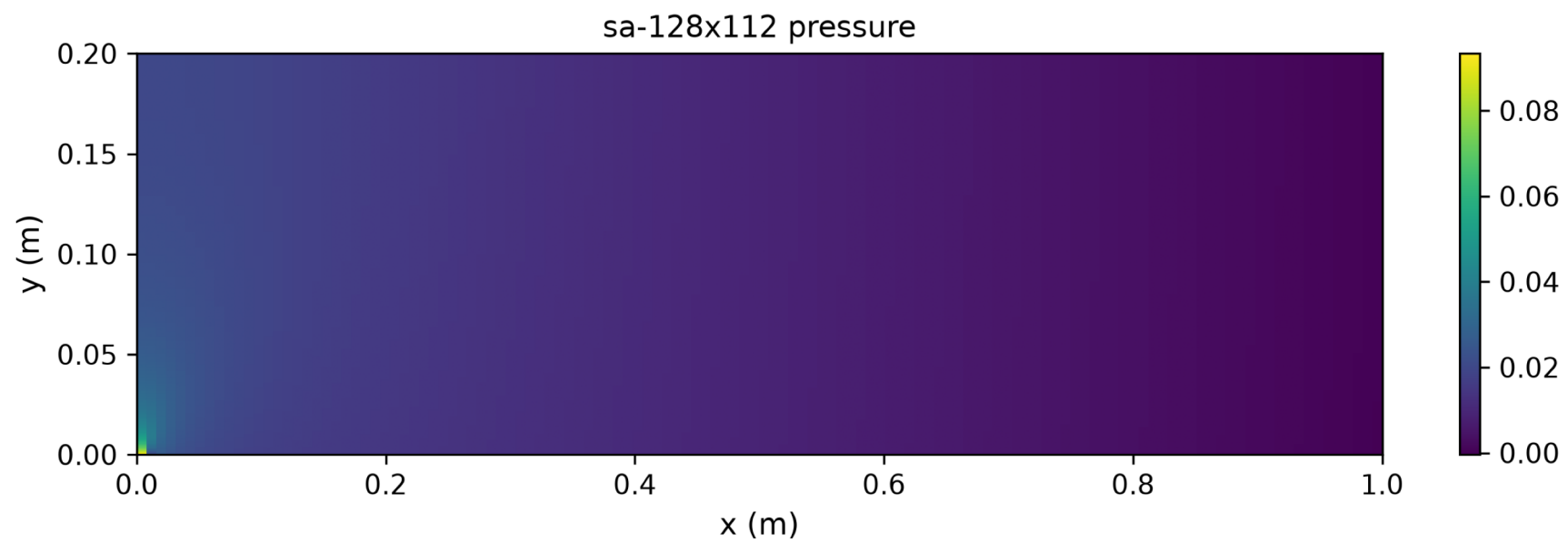
Grid	Converged	Steps	Mean Cf	Empirical error
sa-64x56	True	78	0.00691056	6.6141%
sa-128x112	True	99	0.00696384	5.8940%
sa-256x224	True	132	0.00702561	5.0593%

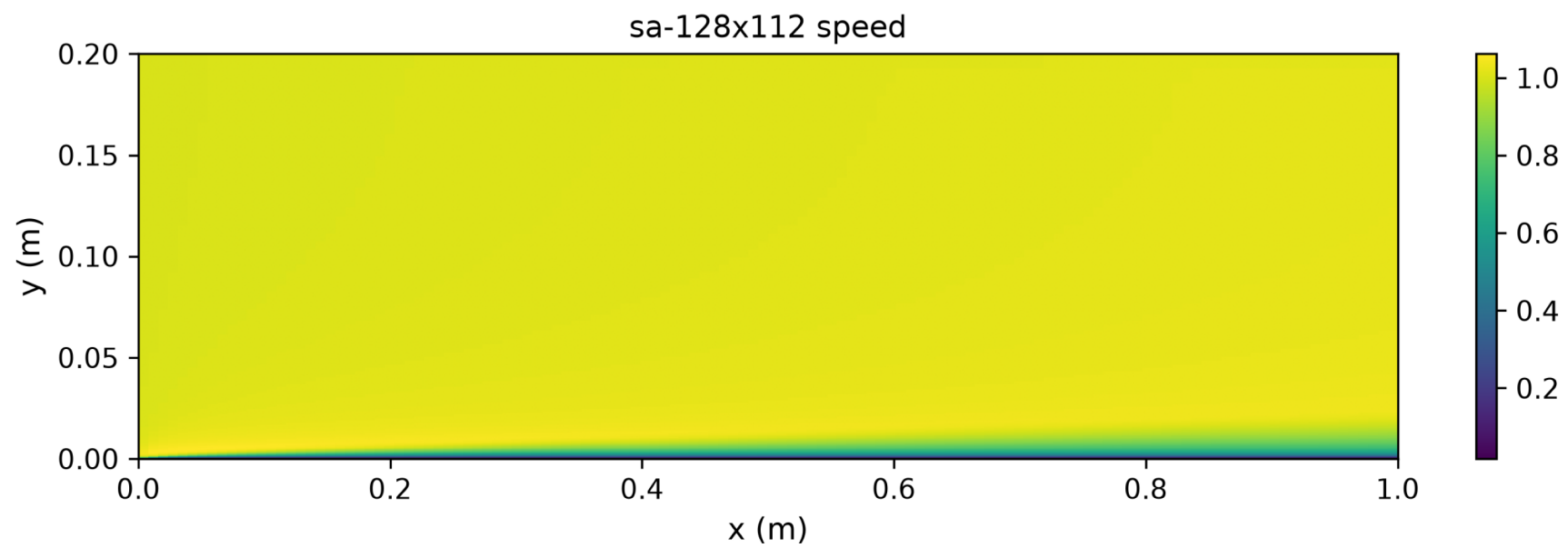
Original SA model, $Re=100000$; actual coarse fields seed refined solves.
Empirical mean Cf is not a matched NASA or experimental validation.
Two CPU levels and one CUDA level; no finest-grid equivalence claim.
All physics and solver gates are reported independently.

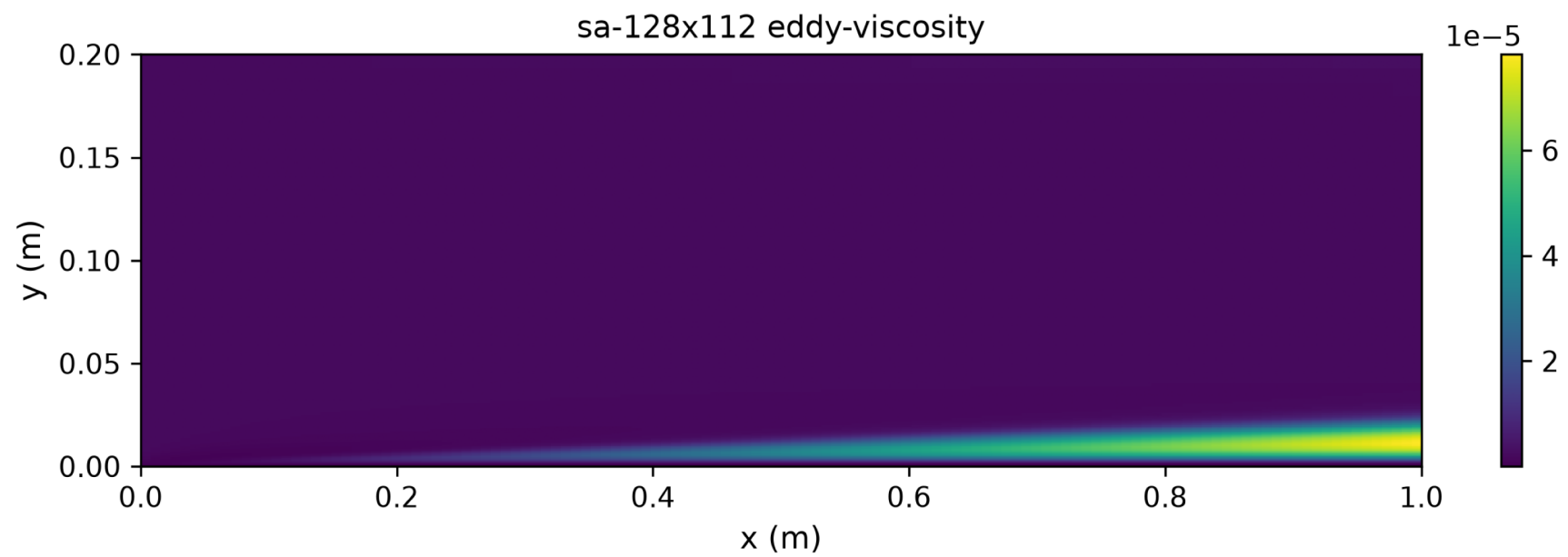


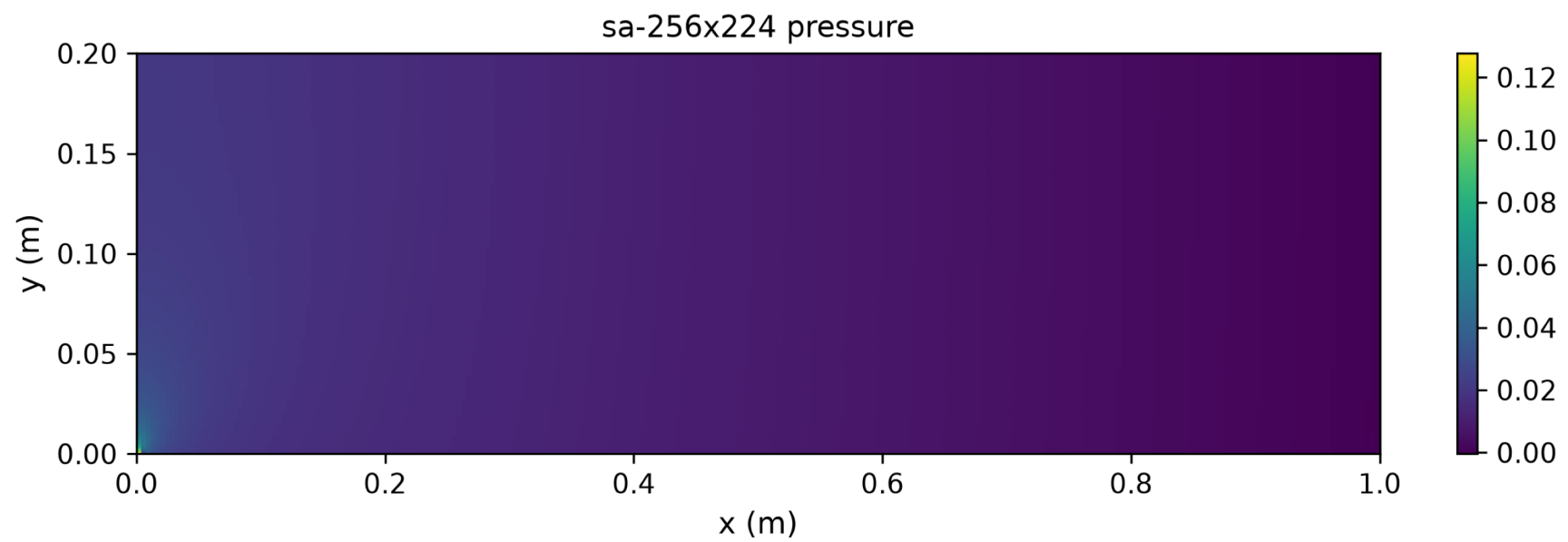


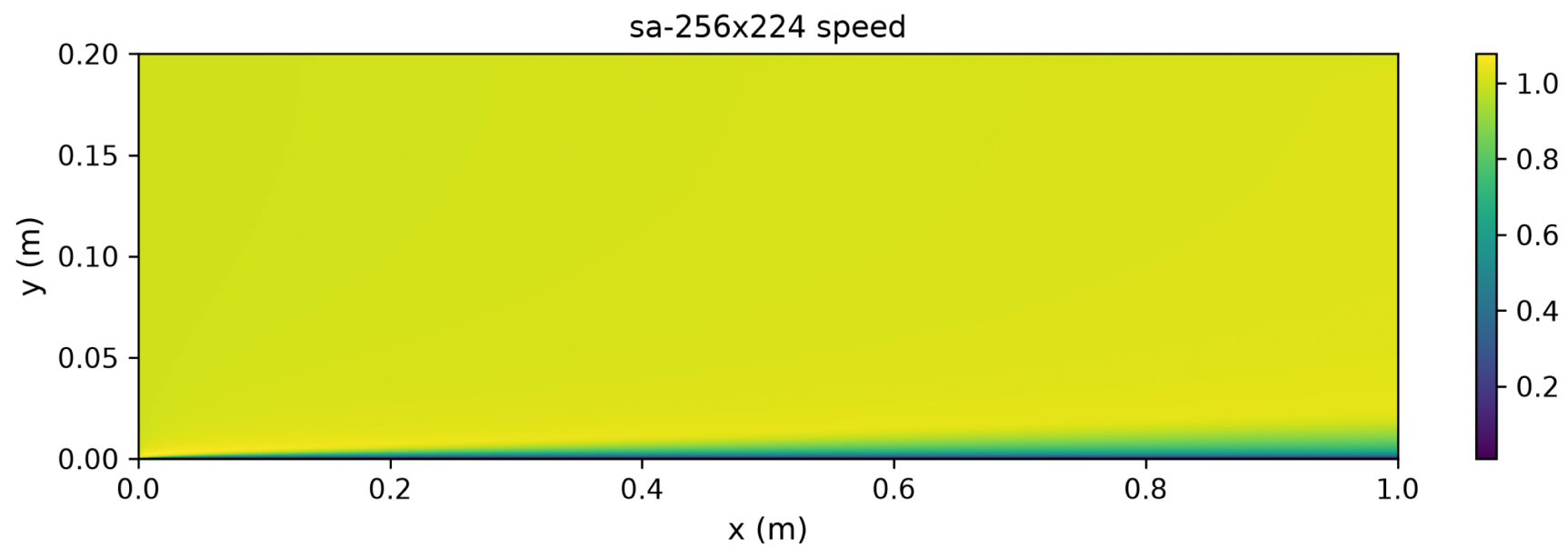


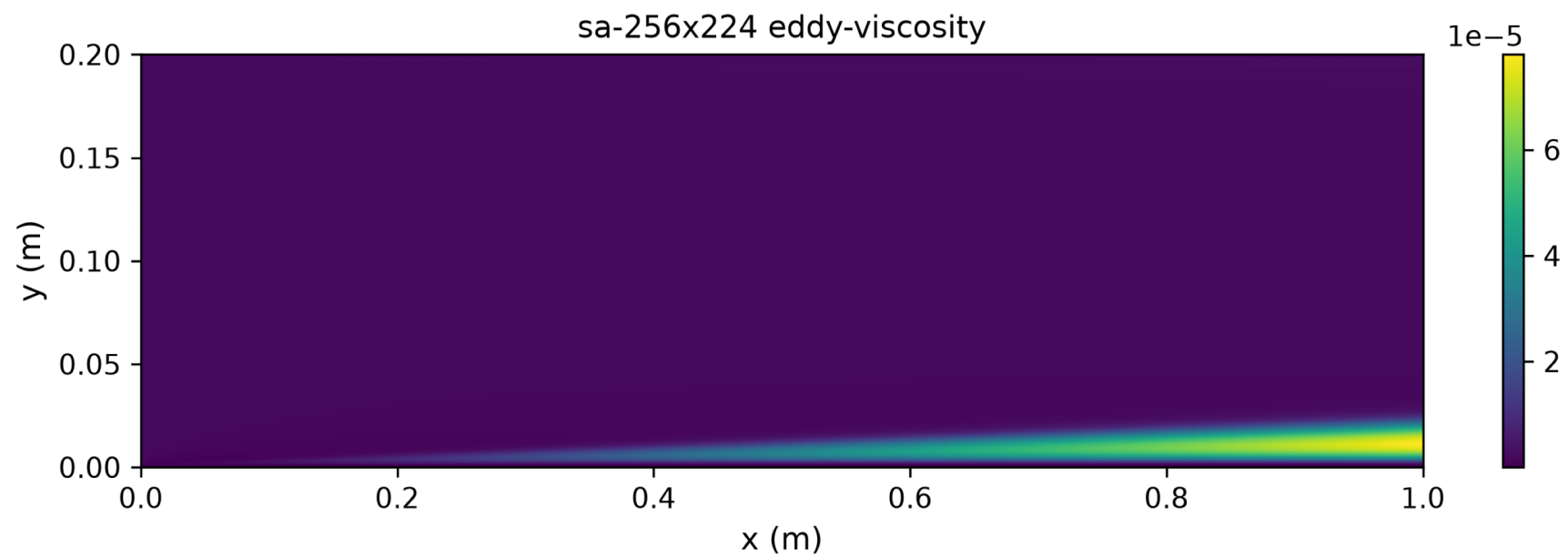


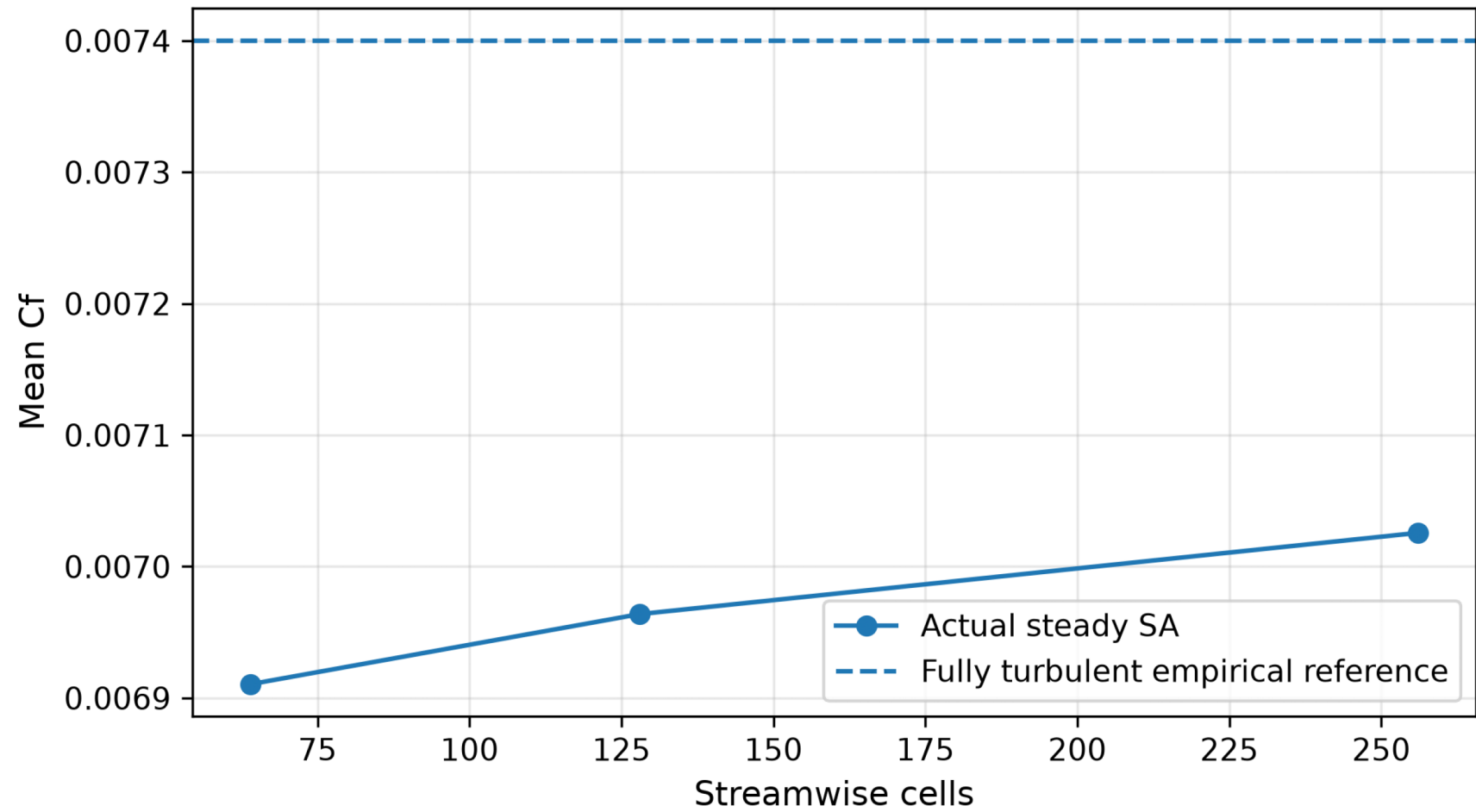




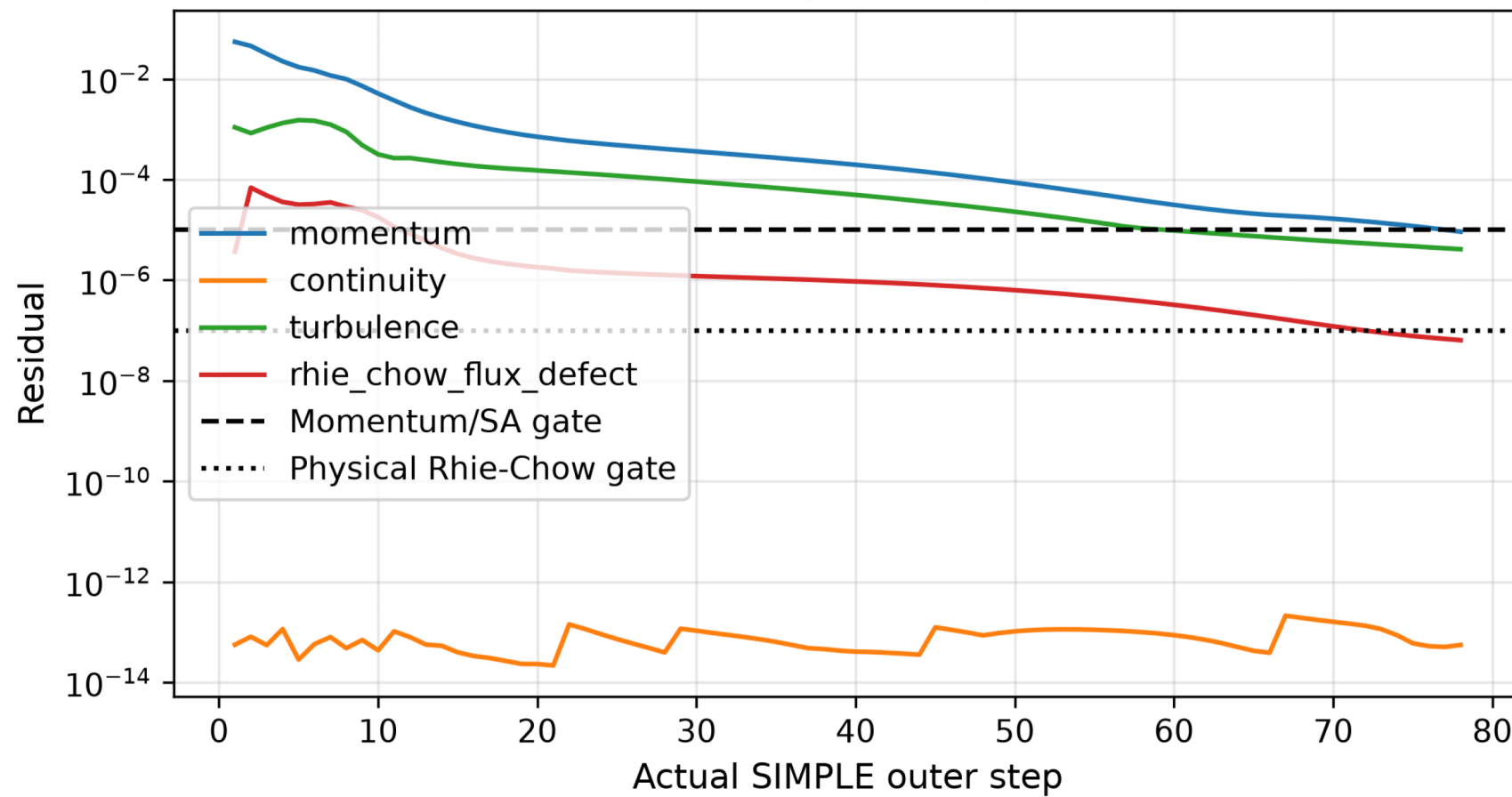








sa-64x56 cpu convergence



sa-128x112 cpu convergence

